KG4Ex-NOV on Eedi | Ablation experiments

Final graph: theta_m=0.8, theta_f=0.8, rec=50; zero-valued exercise-forgetting edges removed

Correct-response history; N = 5, 10, ..., 50; higher is better

	Six ablation models					
N	**2CKG4ER**	**No-mastery**	**No-forgetting**	**No-student feature**	**No-exercise feature**	**ID-based Only**
5	0.982 ± 0.002	0.984 ± 0.003	**0.991 ± 0.001**	0.983 ± 0.001	0.983 ± 0.001	0.983 ± 0.001
10	0.982 ± 0.002	0.983 ± 0.002	**0.989 ± 0.000**	0.983 ± 0.001	0.983 ± 0.001	0.983 ± 0.000
15	0.983 ± 0.002	0.983 ± 0.001	**0.990 ± 0.001**	0.982 ± 0.001	0.984 ± 0.001	0.983 ± 0.000
20	0.983 ± 0.001	0.983 ± 0.001	**0.990 ± 0.001**	0.982 ± 0.001	0.984 ± 0.001	0.983 ± 0.001
25	0.983 ± 0.001	0.983 ± 0.000	**0.990 ± 0.001**	0.982 ± 0.000	0.984 ± 0.001	0.983 ± 0.000
30	0.983 ± 0.001	0.982 ± 0.000	**0.989 ± 0.001**	0.982 ± 0.000	0.984 ± 0.001	0.982 ± 0.000
35	0.983 ± 0.001	0.982 ± 0.000	**0.988 ± 0.001**	0.982 ± 0.000	0.984 ± 0.001	0.982 ± 0.000
40	0.983 ± 0.000	0.982 ± 0.000	**0.987 ± 0.000**	0.982 ± 0.000	0.984 ± 0.000	0.982 ± 0.000
45	0.983 ± 0.000	0.982 ± 0.000	**0.986 ± 0.000**	0.982 ± 0.000	0.983 ± 0.000	0.982 ± 0.000
50	0.983 ± 0.000	0.982 ± 0.000	**0.985 ± 0.000**	0.982 ± 0.000	0.983 ± 0.000	0.982 ± 0.000

KG4Ex-NOV(u,e) = 1 - |K(e) intersection C(u)| / |K(e) union C(u)|; average over Top-N exercises, then test students.

C(u) contains knowledge concepts from correctly answered test-history exercises. Values are mean ± sample SD across seeds 2024, 2025, 2026.

Bold: best; underline: second best, ranked by unrounded mean within this table and N row.

KG4Ex-NOV on Eedi | Comparison experiments

Final graph: theta_m=0.8, theta_f=0.8, rec=50; zero-valued exercise-forgetting edges removed

Correct-response history; N = 5, 10, ..., 50; higher is better

	Main model and eight comparison models								
N	**2CKG4ER**	**EB-CF**	**SB-CF**	**CBF**	**TransE**	**TransE-adv**	**RotatE**	**DistMult**	**ComplEx**
5	**0.982 ± 0.002**	0.969	0.975	0.977	0.976 ± 0.000	0.975 ± 0.001	0.979 ± 0.000	0.974 ± 0.001	0.973 ± 0.002
10	**0.982 ± 0.002**	0.970	0.975	0.976	0.976 ± 0.000	0.975 ± 0.000	0.979 ± 0.001	0.974 ± 0.001	0.973 ± 0.002
15	**0.983 ± 0.002**	0.971	0.976	0.976	0.976 ± 0.000	0.975 ± 0.000	0.979 ± 0.001	0.974 ± 0.001	0.973 ± 0.002
20	**0.983 ± 0.001**	0.971	0.976	0.976	0.975 ± 0.000	0.975 ± 0.001	0.979 ± 0.001	0.974 ± 0.001	0.973 ± 0.001
25	**0.983 ± 0.001**	0.971	0.976	0.976	0.975 ± 0.000	0.975 ± 0.000	0.979 ± 0.001	0.974 ± 0.001	0.974 ± 0.001
30	**0.983 ± 0.001**	0.972	0.976	0.976	0.975 ± 0.000	0.975 ± 0.000	0.979 ± 0.000	0.974 ± 0.001	0.974 ± 0.001
35	**0.983 ± 0.001**	0.972	0.976	0.976	0.975 ± 0.000	0.975 ± 0.001	0.979 ± 0.000	0.974 ± 0.001	0.974 ± 0.001
40	**0.983 ± 0.000**	0.972	0.976	0.976	0.975 ± 0.000	0.975 ± 0.001	0.979 ± 0.000	0.974 ± 0.001	0.974 ± 0.001
45	**0.983 ± 0.000**	0.973	0.976	0.976	0.975 ± 0.000	0.975 ± 0.001	0.979 ± 0.000	0.974 ± 0.001	0.974 ± 0.001
50	**0.983 ± 0.000**	0.973	0.976	0.976	0.975 ± 0.001	0.975 ± 0.001	0.979 ± 0.000	0.974 ± 0.001	0.974 ± 0.001

KG4Ex-NOV(u,e) = 1 - |K(e) intersection C(u)| / |K(e) union C(u)|; average over Top-N exercises, then test students.

C(u) contains knowledge concepts from correctly answered test-history exercises. Values are mean ± sample SD across seeds 2024, 2025, 2026.

EB-CF, SB-CF and CBF are deterministic single runs. Bold: best; underline: second best, ranked by unrounded mean within this table and N row.

KG4Ex-NOV on Algebra2005 | Ablation experiments

Final graph: theta_m=0.8, theta_f=0.8, rec=50; zero-valued exercise-forgetting edges removed

Correct-response history; N = 5, 10, ..., 50; higher is better

	Six ablation models					
N	**2CKG4ER**	**No-mastery**	**No-forgetting**	**No-student feature**	**No-exercise feature**	**ID-based Only**
5	0.986 ± 0.000	0.928 ± 0.007	**0.999 ± 0.000**	0.987 ± 0.002	0.987 ± 0.001	0.987 ± 0.000
10	0.986 ± 0.001	0.931 ± 0.006	**0.999 ± 0.000**	0.986 ± 0.004	0.987 ± 0.001	0.986 ± 0.001
15	0.986 ± 0.002	0.932 ± 0.005	**0.999 ± 0.000**	0.986 ± 0.004	0.987 ± 0.002	0.985 ± 0.001
20	0.988 ± 0.002	0.929 ± 0.006	**0.999 ± 0.001**	0.985 ± 0.004	0.988 ± 0.001	0.984 ± 0.001
25	0.987 ± 0.001	0.928 ± 0.008	**0.998 ± 0.001**	0.984 ± 0.004	0.987 ± 0.001	0.983 ± 0.001
30	0.985 ± 0.001	0.929 ± 0.007	**0.997 ± 0.000**	0.984 ± 0.004	0.986 ± 0.001	0.982 ± 0.001
35	0.985 ± 0.001	0.927 ± 0.008	**0.997 ± 0.000**	0.983 ± 0.004	0.985 ± 0.001	0.982 ± 0.001
40	0.985 ± 0.001	0.925 ± 0.008	**0.996 ± 0.000**	0.982 ± 0.004	0.985 ± 0.000	0.981 ± 0.001
45	0.985 ± 0.000	0.922 ± 0.007	**0.996 ± 0.000**	0.981 ± 0.004	0.985 ± 0.000	0.980 ± 0.002
50	0.985 ± 0.001	0.920 ± 0.007	**0.996 ± 0.001**	0.980 ± 0.004	0.986 ± 0.001	0.979 ± 0.002

KG4Ex-NOV(u,e) = 1 - |K(e) intersection C(u)| / |K(e) union C(u)|; average over Top-N exercises, then test students.

C(u) contains knowledge concepts from correctly answered test-history exercises. Values are mean ± sample SD across seeds 2024, 2025, 2026.

Bold: best; underline: second best, ranked by unrounded mean within this table and N row.

KG4Ex-NOV on Algebra2005 | Comparison experiments

Final graph: theta_m=0.8, theta_f=0.8, rec=50; zero-valued exercise-forgetting edges removed

Correct-response history; N = 5, 10, ..., 50; higher is better

	Main model and eight comparison models								
N	**2CKG4ER**	**EB-CF**	**SB-CF**	**CBF**	**TransE**	**TransE-adv**	**RotatE**	**DistMult**	**ComplEx**
5	**0.986 ± 0.000**	0.738	0.888	0.964	0.981 ± 0.001	0.986 ± 0.001	0.986 ± 0.000	0.908 ± 0.005	0.863 ± 0.008
10	**0.986 ± 0.001**	0.746	0.922	0.962	0.981 ± 0.001	0.986 ± 0.001	0.985 ± 0.001	0.901 ± 0.006	0.860 ± 0.005
15	**0.986 ± 0.002**	0.749	0.929	0.958	0.981 ± 0.001	0.985 ± 0.001	0.986 ± 0.001	0.895 ± 0.007	0.856 ± 0.004
20	**0.988 ± 0.002**	0.750	0.927	0.954	0.979 ± 0.001	0.985 ± 0.001	0.985 ± 0.002	0.893 ± 0.007	0.852 ± 0.003
25	**0.987 ± 0.001**	0.753	0.922	0.952	0.978 ± 0.002	0.985 ± 0.001	0.984 ± 0.002	0.891 ± 0.006	0.849 ± 0.002
30	**0.985 ± 0.001**	0.755	0.920	0.949	0.977 ± 0.003	0.985 ± 0.001	0.983 ± 0.002	0.889 ± 0.006	0.847 ± 0.001
35	**0.985 ± 0.001**	0.757	0.918	0.947	0.976 ± 0.003	0.985 ± 0.002	0.983 ± 0.002	0.888 ± 0.006	0.845 ± 0.001
40	**0.985 ± 0.001**	0.757	0.916	0.946	0.975 ± 0.003	0.984 ± 0.002	0.983 ± 0.002	0.886 ± 0.007	0.844 ± 0.001
45	**0.985 ± 0.000**	0.758	0.915	0.945	0.974 ± 0.003	0.983 ± 0.002	0.982 ± 0.002	0.885 ± 0.006	0.843 ± 0.001
50	**0.985 ± 0.001**	0.760	0.915	0.943	0.973 ± 0.002	0.982 ± 0.002	0.980 ± 0.002	0.884 ± 0.006	0.842 ± 0.001

KG4Ex-NOV(u,e) = 1 - |K(e) intersection C(u)| / |K(e) union C(u)|; average over Top-N exercises, then test students.

C(u) contains knowledge concepts from correctly answered test-history exercises. Values are mean ± sample SD across seeds 2024, 2025, 2026.

EB-CF, SB-CF and CBF are deterministic single runs. Bold: best; underline: second best, ranked by unrounded mean within this table and N row.

KG4Ex-NOV on XES3G5M-sub-small | Ablation experiments

Final graph: theta_m=0.8, theta_f=0.8, rec=50; zero-valued exercise-forgetting edges removed

Correct-response history; N = 5, 10, ..., 50; higher is better

	Six ablation models					
N	**2CKG4ER**	**No-mastery**	**No-forgetting**	**No-student feature**	**No-exercise feature**	**ID-based Only**
5	0.994 ± 0.000	0.996 ± 0.000	**0.998 ± 0.000**	0.995 ± 0.000	0.995 ± 0.000	0.995 ± 0.000
10	0.995 ± 0.000	0.996 ± 0.000	**0.997 ± 0.000**	0.995 ± 0.000	0.995 ± 0.000	0.995 ± 0.000
15	0.995 ± 0.000	0.996 ± 0.000	**0.997 ± 0.000**	0.995 ± 0.000	0.995 ± 0.000	0.995 ± 0.000
20	0.995 ± 0.000	0.996 ± 0.000	**0.997 ± 0.000**	0.995 ± 0.000	0.995 ± 0.000	0.996 ± 0.000
25	0.995 ± 0.000	0.996 ± 0.000	**0.997 ± 0.000**	0.995 ± 0.000	0.995 ± 0.000	0.996 ± 0.000
30	0.994 ± 0.000	0.996 ± 0.000	**0.997 ± 0.000**	0.995 ± 0.000	0.995 ± 0.000	0.996 ± 0.000
35	0.995 ± 0.000	0.996 ± 0.000	**0.997 ± 0.000**	0.995 ± 0.000	0.995 ± 0.000	0.996 ± 0.000
40	0.995 ± 0.000	0.996 ± 0.000	**0.997 ± 0.000**	0.995 ± 0.000	0.995 ± 0.000	0.996 ± 0.000
45	0.995 ± 0.000	0.996 ± 0.000	**0.996 ± 0.000**	0.995 ± 0.000	0.995 ± 0.000	0.996 ± 0.000
50	0.995 ± 0.000	0.995 ± 0.000	**0.996 ± 0.000**	0.995 ± 0.000	0.995 ± 0.000	0.996 ± 0.000

KG4Ex-NOV(u,e) = 1 - |K(e) intersection C(u)| / |K(e) union C(u)|; average over Top-N exercises, then test students.

C(u) contains knowledge concepts from correctly answered test-history exercises. Values are mean ± sample SD across seeds 2024, 2025, 2026.

Bold: best; underline: second best, ranked by unrounded mean within this table and N row.

KG4Ex-NOV on XES3G5M-sub-small | Comparison experiments

Final graph: theta_m=0.8, theta_f=0.8, rec=50; zero-valued exercise-forgetting edges removed

Correct-response history; N = 5, 10, ..., 50; higher is better

	Main model and eight comparison models								
N	**2CKG4ER**	**EB-CF**	**SB-CF**	**CBF**	**TransE**	**TransE-adv**	**RotatE**	**DistMult**	**ComplEx**
5	0.994 ± 0.000	0.991	0.994	**0.995**	0.993 ± 0.000	0.993 ± 0.000	0.991 ± 0.000	0.993 ± 0.000	0.993 ± 0.000
10	**0.995 ± 0.000**	0.991	0.994	0.994	0.993 ± 0.000	0.993 ± 0.000	0.992 ± 0.000	0.993 ± 0.000	0.993 ± 0.000
15	**0.995 ± 0.000**	0.991	0.994	0.994	0.993 ± 0.000	0.993 ± 0.000	0.992 ± 0.000	0.993 ± 0.000	0.993 ± 0.000
20	**0.995 ± 0.000**	0.991	0.994	0.994	0.993 ± 0.000	0.993 ± 0.000	0.992 ± 0.000	0.993 ± 0.000	0.993 ± 0.000
25	**0.995 ± 0.000**	0.991	0.994	0.994	0.993 ± 0.000	0.993 ± 0.000	0.992 ± 0.000	0.993 ± 0.000	0.993 ± 0.000
30	**0.994 ± 0.000**	0.991	0.994	0.994	0.993 ± 0.000	0.993 ± 0.000	0.992 ± 0.000	0.993 ± 0.000	0.993 ± 0.000
35	**0.995 ± 0.000**	0.991	0.994	0.994	0.993 ± 0.000	0.993 ± 0.000	0.993 ± 0.000	0.993 ± 0.000	0.993 ± 0.000
40	**0.995 ± 0.000**	0.991	0.994	0.994	0.993 ± 0.000	0.993 ± 0.000	0.993 ± 0.000	0.993 ± 0.000	0.993 ± 0.000
45	**0.995 ± 0.000**	0.991	0.994	0.994	0.993 ± 0.000	0.993 ± 0.000	0.993 ± 0.000	0.993 ± 0.000	0.993 ± 0.000
50	**0.995 ± 0.000**	0.991	0.994	0.994	0.993 ± 0.000	0.993 ± 0.000	0.993 ± 0.000	0.993 ± 0.000	0.993 ± 0.000

KG4Ex-NOV(u,e) = 1 - |K(e) intersection C(u)| / |K(e) union C(u)|; average over Top-N exercises, then test students.

C(u) contains knowledge concepts from correctly answered test-history exercises. Values are mean ± sample SD across seeds 2024, 2025, 2026.

EB-CF, SB-CF and CBF are deterministic single runs. Bold: best; underline: second best, ranked by unrounded mean within this table and N row.